LECTURE 13:
MARKOV CHAINS

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

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Definition of Markov chains

Our first example of a martingale was a simple random walk $S_n = \sum_{i=1}^n X_i$, where X_i are i.i.d. random variables. In the "random walk" interpretation, the question of whether the expectation is unchanged with time is not all we could care about. Some other questions we might be interested in:

(1) Do we return to the point where we started?
(2) Do we return infinitely many times or stop returning and drift away eventually?
(3) Do we expect to visit every point and how soon?
(4) In the long run, where do we expect to be? Distribution?

Symmetry is optional here. Another key property: *the next move (S_{n+1}) only depends on the current position (S_n), and not on the past ($S_0, S_1, \dots, S_{n-1}$).*

Definition 13.1. Given a filtration $\{F_n\}$, a $\mathcal{F}_n$-adapted stochastic process $\{X_n\}$ with $X_n : (\Omega, \mathcal{F}_n, \mathbb{P}) \to (S, \mathcal{S})$ is a **Markov chain** (MC) if for all n and any $A \in \mathcal{S}$,

$$\mathbb{P}(X_{n+1} \in A \mid \mathcal{F}_n) = \mathbb{P}(X_{n+1} \in A \mid X_n).$$

Note that the conditional probability of an event is just a special case of conditional expectation: $\mathbb{P}(\text{Event} \mid \mathcal{F}_n) = \mathbb{E}(\mathbb{1}_{\text{Event}} \mid \mathcal{F}_n)$. If the state space $(S, \mathcal{S})$ is discrete, then the definition above is equivalent to the following: $(X_n)_n$ is a Markov chain if for all n and $a_0, a_1, \dots, a_n, a_{n+1} \in S$,

$$\mathbb{P}(X_{n+1} = a_{n+1} \mid X_0 = a_0, \dots, X_n = a_n) = \mathbb{P}(X_{n+1} = a_{n+1} \mid X_n = a_n).$$

This definition significantly simplifies computations. If we use the notation $\bar{X}_0^n = (X_0, \dots, X_n)$ and $\bar{s}_0^n = (s_0, \dots, s_n)$, then

$$\begin{aligned}\mathbb{P}(\bar{X}_0^n = \bar{s}_0^n) &= \mathbb{P}(X_0 = s_0) \cdot \prod_{i=1}^n \mathbb{P}(X_i = s_i \mid \bar{X}_0^{i-1} = \bar{s}_0^{i-1}) \\ &= \mathbb{P}(X_0 = s_0) \cdot \prod_{i=1}^n \mathbb{P}(X_i = s_i \mid X_{i-1} = s_{i-1})\end{aligned}$$

Here, $\mathbb{P}(X_i = s_i \mid X_{i-1} = s_{i-1})$ are the **transition probabilities**. These may be convenient described using a **transition matrix** $P_{xy}(j) = \mathbb{P}(X_j = y \mid X_{j-1} = x)$, which may possibly depend on the time j. If $P = P_{xy}$ does not depend on the time j, then we say that $(X_n)_n$

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is a **time-homogeneous Markov chain**. In this case, the Markov chain is defined by its transition matrix P, and we would simply have

$$\mathbb{P}(\bar{X}_0^n = \bar{s}_0^n) = \underbrace{\mathbb{P}(X_0 = s_0)}_{\text{initial distribution}} \cdot \underbrace{\prod_{i=1}^{n} P_{a_{i-1},a_i}}_{\text{transitions}}.$$

Theorem 13.2 (Chapman-Kolmogorov equations)**.** *Let $(X_n)_n$ be a (discrete state space, time-homogeneous) Markov chain with transition matrix P. Then for all $n \geq 1$ and $x, y \in S$,*

$$P_{xy}^{(n)} := \mathbb{P}(X_n = y \mid X_0 = x) = (P^n)_{xy}.$$

Proof. This can be shown by induction. The base case for $n = 1$ is true by definition. Assuming the induction hypothesis holds for $n - 1$, we have

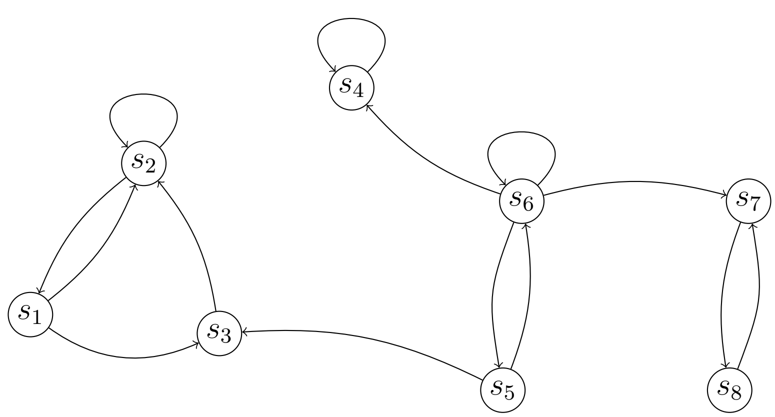
$$\begin{aligned}
\mathbb{P}(X_n = y \mid X_0 = x) &= \sum_{z \in S} \mathbb{P}(X_n = y, X_{n-1} = z \mid X_0 = x) \\
&= \sum_{z \in S} \mathbb{P}(X_n = y \mid X_{n-1} = z, X_0 = x) \cdot \mathbb{P}(X_{n-1} = z \mid X_0 = x) \\
&= \sum_{z \in S} \mathbb{P}(X_n = y \mid X_{n-1} = z) \cdot (P^{n-1})_{xz} \\
&= \sum_{z \in S} P_{zy} \cdot (P^{n-1})_{xz} = (P^n)_{xy}.
\end{aligned}$$

□

Example 13.3 (Sampling without replacement)**.** Let S be a set with $|S| = n$. Let $X_1 \sim \text{Unif}(S)$, and $X_{j+1} \sim \text{Unif}(S \setminus \{X_1, \ldots, X_j\})$. Then X_j is not a Markov chain! But $A_j = \{X_1, \ldots, X_j\}$ is a time-homogeneous Markov chain on 2^S:

$$\mathbb{P}(B \mid A) = \begin{cases} \frac{1}{n-|A|} & \text{if } |B \setminus A| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

A Markov chain of any nature can be represented by a directed graph: e.g.,



Our goal: *to understand long-term behavior of a Markov chain.*

Recurrence and transience

In particular, let us consider our first example with $S_n = \sum_{i=1}^n X_i$ again. Will the random walk ever return to the starting point?

Consider the *hitting time*

$$\tau_s := \inf\{n \geq 1 : X_n = s\},$$

which is the first time that the process hits the state s. More generally,

$$\tau_s^k := \inf\{n > \tau_s^{k-1} : X_n = s\}$$

defines the kth hitting time. The event that $(X_n)_n$ hits s in finite time is equivalent to the event $\{\tau_s < \infty\}$.

Theorem 13.4. *Let $(X_n)_n$ be a Markov chain with transition matrix P and initial state $X_0 = s$. Then X_n returns to s infinitely many times with probability one if and only if*

$$\sum_{n=1}^{\infty} P_{ss}^{(n)} = \infty \iff \mathbb{P}(\tau_s < \infty \mid X_0 = s) = 1.$$

In this case, we say that s is a ***recurrent state****. Furthermore, X_n returns to s finitely many times with probability one (i.e., it eventually drifts away and never returns) if and only if*

$$\sum_{n=1}^{\infty} P_{ss}^{(n)} < \infty \iff \mathbb{P}(\tau_s < \infty \mid X_0 = s) < 1.$$

In this case, we say that s is a ***transient state****.*

Note: we can further classify a recurrent state s as ***positive recurrent*** if $\mathbb{E}_s[\tau_s] = \mathbb{E}[\tau_s|X_0 = s] < \infty$ or ***null-recurrent*** if $\mathbb{E}_s[\tau_s] = \mathbb{E}[\tau_s|X_0 = s] = \infty$.[1]

Proof. Let N count the number of times the Markov chain returns to s. Then

$$\sum_{n=1}^{\infty} P_{ss}^{(n)} = \sum_{n=1}^{\infty} \mathbb{P}(X_n = s \mid X_0 = s) = \sum_{n=1}^{\infty} \mathbb{E}(\mathbb{1}_{X_n = s} \mid X_0 = s) = \mathbb{E}_s[N].$$

We claim that $\mathbb{P}_s(N \geq k) = \rho^k$, where $\rho := \mathbb{P}_s(\tau_s < \infty)$. Note that having this claim is enough: indeed,

$$\mathbb{E}_s[N] = \sum_{k=1}^{\infty} \mathbb{P}_s(N \geq k) = \sum_{k=1}^{\infty} \rho^k = \infty$$

if and only if $\rho = 1$, which is also equivalent to $N = \infty$ a.s. Conversely, the $\mathbb{E}_s[N] < \infty$ is equivalent to $\rho < 1$ or $N < \infty$ a.s.

We will now prove the claim by induction. It is clear for $k = 1$ by definition. The rest of the proof is based on the so-called **strong Markov property**: informally, we can restart a Markov chain at a stopping time and the future will only depend on the position at the stopping time (the Markov property used to define the chain states that this is true for a fixed stopping time $T = n$ a.s.).

[1] Throughout the note, we denote by $\mathbb{E}_s$ and $\mathbb{P}_s$ the expectation and probability conditioned on the event $\{X_0 = s\}$ that a MC starts from the state s. Similarly, $\mathbb{E}_\pi$ and $\mathbb{P}_\pi$ are conditional on the random variable X_0 that has distribution π (for every $s \in S$, we know that $X_0 = s$ with probability $\pi(s)$).

However, our argument is explicit. We compute:

$$\begin{aligned}
\mathbb{P}_s(N \geq k) &= \mathbb{P}_s(\tau_s^k < \infty) \\
&= \sum_i \mathbb{P}_s(\tau_s^{k-1} = i, \, \exists j > i : X_j = s) \\
&= \sum_i \mathbb{P}(\exists j > i : X_j = s \mid \tau_s^{k-1} = i, X_0 = s) \cdot \mathbb{P}(\tau_s^{k-1} = i \mid X_0 = s) \\
&= \sum_i \mathbb{P}(\exists j > i : X_j = s \mid X_i = s) \cdot \mathbb{P}(\tau_s^{k-1} = i \mid X_0 = s).
\end{aligned}$$

Here we used the fact that $\{\tau_s^{k-1} = i, X_0 = s\} \in \mathcal{F}_i$ and the usual Markov property. Hence,

$$\mathbb{P}_s(N \geq k) = \rho \sum_i \mathbb{P}(\tau_s^{k-1} = i \mid X_0 = s) = \rho \mathbb{P}_s(\tau_s^{k-1} < \infty).$$

By the induction hypothesis, this implies $\mathbb{P}_s(N \geq k) = \rho \cdot \rho^{k-1} = \rho^k$. $\square$

Does information about one state give us information about all states?

Definition 13.5. Let P be a transition matrix. We say that a state s is **accessible** from state t (written $t \to s$) if $\exists 1 \leq n < \infty$ such that $P_{ts}^{(n)} > 0$. If s is accessible from t and t is accessible from t, then we say that s and t **communicate** (written $s \leftrightarrow t$). If every state is accessible from every other state (i.e., $s \leftrightarrow t$ for all s, t), we say that the Markov chain is **irreducible**.

Theorem 13.6. *If s and t communicate ($s \leftrightarrow t$), then they are both recurrent or both transient.*

Proof. Suppose that $P_{st}^{(n)} > 0$ and $P_{ts}^{(m)} > 0$. Then for $\ell \geq n + m$,

$$P_{ss}^{(\ell)} = (P^\ell)_{ss} = (P^n P^{\ell-n-m} P^m)_{ss} \geq (P^n)_{st} (P^{\ell-n-m})_{tt} (P^m)_{ts} = P_{st}^{(n)} P_{tt}^{(\ell-n-m)} P_{ts}^{(m)}.$$

Here, the inequality follows from the fact $(AB)_{ij} \geq A_{ik} B_{kj}$ for any k since all the matrices in question have non-negative entries. Thus,

$$\sum_{\ell \geq 1} P_{ss}^{(\ell)} \geq \sum_{\ell \geq n+m} P_{ss}^{(\ell)} \geq P_{st}^{(n)} \left(\sum_{\ell \geq n+m} P_{tt}^{(\ell-n-m)} \right) P_{ts}^{(m)}.$$

If t is recurrent, then both sides are ∞, which implies that s is also recurrent. The opposite side follows from a similar argument. $\square$

Theorem 13.7 (Pólya's recurrence theorem)**.** *The simple symmetric random walk on $\mathbb{Z}^d$ is recurrent if $d = 1, 2$, and transient if $d \geq 3$.*

("The drunk person always returns home, but a drunk bird might not.")

Proof. **(d = 1)** The random walk can be represented as $S_n = \sum_{i=1}^n X_i$, where X_i are i.i.d. ± 1 Rademacher random variables. Let R_n be the number of steps to the right (out of n total steps). Then $R_n \sim \text{Binomial}(n, 1/2)$ since each direction is chosen independently with probability $1/2$. We have

$$\mathbb{P}(S_{2n} = 0) = \mathbb{P}(R_{2n} = n) = \binom{2n}{n} \left(\frac{1}{2} \right)^{2n} \sim \frac{1}{\sqrt{\pi n}},$$

which can be shown using Stirling's formula $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$. Thus,

$$\sum_{n=1}^{\infty} \mathbb{P}(S_{2n}=0) \sim \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \infty.$$

So, 0 is a recurrent state by Theorem 13.4.

Now, what is the expected return time to the origin? It is

$$\sum_t t\mathbb{P}(\text{ return in } t \text{ steps}) = \sum_{k=0}^{\infty} 2k\frac{C}{\sqrt{k}} = \infty$$

(d = 2) Here we need to count steps in both x and y directions, which are almost independent. The following trick reduces the problem to two independent copies of the one-dimensional random walk: rotate the grid by 45 degrees and rescale by $\sqrt{2}$. Then each step changes by $(+1,+1)$, $(+1,-1)$, $(-1,+1)$, or $(+1,+1)$ with probabilities given by two independent one-dimensional symmetric random walk. Thus,

$$\mathbb{P}(S_{2n}=0) \sim \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

which shows that the origin is also a recurrent state.

(d > 2) The number of steps in each direction is $\sim$ Binomial$(n, 1/d)$. Given k steps in a fixed direction, the number of positive steps is $\sim$ Binomial$(k, 1/2)$. So,

$$\mathbb{P}(k/2 \text{ positive steps in direction } i \mid k \text{ steps in direction } i) = \begin{cases} 0 & \text{if } k \text{ is odd,} \\ \sim \frac{1}{\sqrt{k}} & \text{if } k \text{ is even.} \end{cases}$$

This is a good estimate if k is large (e.g., $k = O(n)$), and one can show that small k (e.g., $k < n/d$) is unlikely. Overall, it can be shown that

$$\begin{aligned}\mathbb{P}(S_{2n}=0) &= \sum_{\substack{k_1,\ldots,k_d \\ k_1+\ldots+k_d=2n}} \prod_i \mathbb{P}(k_i/2 \text{ positive steps} \mid k_i \text{ steps in dir. } i)\cdot \mathbb{P}(k_i \text{ steps in dir. } i) \\ &\le de^{-cn} + cn^{-d/2}.\end{aligned}$$

Hence,

$$\sum_{n=1}^{\infty} \mathbb{P}(S_{2n}=0) \le \sum_{n=1}^{\infty} (de^{-cn} + cn^{-d/2}) < \infty$$

as long as $d > 2$, which shows that the origin is transient. □

Stationary distributions

So far, we were interested in the following long-term behavior: starting from a state s, how many times do we return to s?

Some more refined questions include: how frequently will we be returning to each state? What is the distribution of being in each state (in the long run)? How does it depend on where we start?

We will answer these questions in the more restricted case of a Markov chain with *finite state space* S. First, what can the limiting distribution on S be?

Definition 13.8. A **distribution** on a finite state space S with $|S| = N$ can be identified with a (row) vector $\mu = (\mu_i)_{i=1}^N$ such that $\mu_i \geq 0$ and $\sum_{i=1}^N \mu_i = 1$.

We say that μ is a **stationary distribution** for the transition matrix P if $\mu P = \mu$, i.e.,

$$\sum_{i=1}^N \mu_i P_{ij} = \mu_j \quad \text{for all } j.$$

What does this mean? If μ is a stationary distribution, then $X_0 \sim \mu$ (i.e., $\mathbb{P}(X_0 = i) = \mu_i$) implies that $X_n \sim \mu$ for all $n \geq 1$. Indeed, by the Chapman-Kolmogorov equations

$$\mu_j^{(n)} = \mathbb{P}(X_n = j) = \sum_{i \in S} \mathbb{P}(X_n = j \mid X_0 = i) \cdot \mathbb{P}(X_0 = i) = \sum_{i \in S} \mu_i (P^n)_{ij} = (\mu P^n)_j.$$

So, $\mu^{(n)} = \mu P^n = \mu$ (as $\mu P = \mu$) for any n.

Hence, a stationary distribution is something that can happen in the long run (and remain). This motivates the following questions:

(1) Existence?
(2) Uniqueness?
(3) Convergence?
   - How quickly do we converge to a stationary distribution?
   - Does it depend on the starting point?

Existence.

Theorem 13.9. *Let $(X_n)_n$ be a time-homogeneous Markov chain on a finite state space S. Then $(X_n)_n$ has a stationary distribution.*

Proof. Let P be the transition matrix of $(X_n)_n$. We have shown above that for any distribution μ_0, if $X_0 \sim \mu_0$, then $X_k \sim \mu_0 P^k$. Define the distribution

$$\mu_n := \frac{\mu_0 + \mu_0 P + \ldots + \mu_0 P^n}{n+1}.$$

Note that $(\mu_n)_n$ is a bounded sequence (each element is an average of probability measures). Therefore, there exists a convergent subsequence $(\mu_{n_k})_k$ such that $\mu_{n_k} \to \mu$ as $k \to \infty$. Note that the limit μ must also be a distribution. Furthermore,

$$\mu_{n_k} - \mu_{n_k} P = \frac{\mu_0 - \mu_0 P^{n+1}}{n+1} \to 0 \quad \text{as } k \to \infty,$$

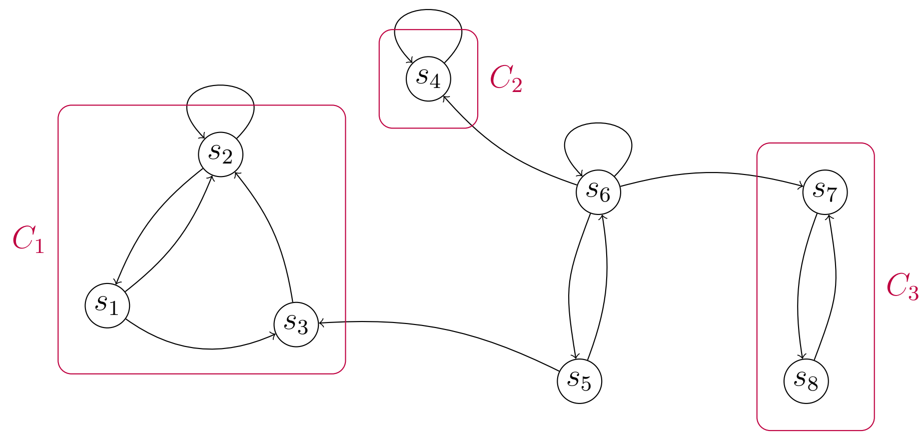
so $\mu = \mu P$. Hence, μ is a stationary distribution for P. $\square$

Uniqueness and convergence. Recall that any Markov chain has a directed graph representation.

Definition 13.10. We say that $S' \subseteq S$ is an **irreducible component** if it is **closed** ($\forall a \notin S', b \in S'$: $P_{ba}^n = 0$ for all $n \geq 1$, i.e., $b \not\to a$) and **irreducible** ($\forall a, b \in S' : P_{ba}^n > 0$ for some $n \geq 1$, i.e., $a \leftrightarrow b$).

We say that $s \in S$ is an **inessential state** if s is not part of any irreducible component.

Example. The irreducible components for the Markov chain with directed graph representation below are $C_1 = \{s_1, s_2, s_3\}$, $C_2 = \{s_4\}$, and $C_3 = \{s_7, s_8\}$. The inessential states $D = \{s_5, s_6\}$. Note that the state space can be partitioned as $S = C_1 \sqcup C_2 \sqcup C_3 \sqcup D$.



Proposition 13.11. *For a Markov chain on a finite state space S,*

$$S = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_r \sqcup D,$$

where $C_1, \ldots, C_r$ are irreducible components (with $r \geq 1$) and D consists of inessential states.

Proof idea. If x is essential, then x is in an irreducible component, say C_1. We can then collect all the elements in S that are also in C_1 (noting that communication $\leftrightarrow$ is an equivalence relationship), and continue the argument on $S \setminus C_1$. $\square$

Observe that a stationary distribution

- cannot have positive probability on an inessential state (exercise: such states are transient!)
- depends on which irreducible component we started in (or get into first if started from an inessential state) – we cannot escape an irreducible component once we are in it (closedness).

The most interesting part is the non-trivial stationary distribution on the states within any one irreducible component. So, from now on, we will focus on irreducible Markov chains (i.e., with a single irreducible component and no inessential states).

Let's return to the question: for the long-term behavior, does it matter where we started (for irreducible Markov chains)? The answer can still be yes.

Definition 13.12. We say that the **period** of a state $x \in S$ is $\gcd\{n \geq 1 : P^n_{xx} > 0\}$. We say that x is **aperiodic** if its period is one.

Claim: if $x \leftrightarrow y$, then x and y have the same period (exercise!) Thus, the period of an irreducible component is well-defined.

It turns out periodicity is the only caveat for the convergence to the unique stationary distribution.

Theorem 13.13. *Let $(X_n)_n$ be an irreducible, aperiodic Markov chain with transition matrix P and stationary distribution π. Then*

(1) *π is unique*

(2) $\exists 0 < \varepsilon < 1$, $C < \infty$ *such that for all* $x \in S$,

$$\sum_{y \in S} |\mathbb{P}(X_n = y \mid X_0 = x) - \pi(y)| \le C(1-\varepsilon)^n.$$

Remark 13.14. (1) The **total variation distance** between distributions two μ, ν is defined as

$$\|\mu - \nu\|_{\mathrm{TV}} := \frac{1}{2} \sum_{y \in S} |\mu(y) - \nu(y)| = \sup_{A \subseteq S} |\mu(A) - \nu(A)|.$$

Thus, the second part of the theorem states that the total variation distance between the n-step distribution, started from x, μ_x^n, and the stationary distribution π decays at an exponential rate (independent of the starting state x!). Moreover, convergence in total variation distribution implies convergence in distribution.

- **Example.** What is the total variation distance between a uniformly random shuffle of a normal deck of 52 cards, and a random shuffle that always keeps the ace of spaces at the bottom?

(2) The second part of the theorem implies the first. However, uniqueness also holds more generally for irreducible Markov chains (not necessarily aperiodic!)

Proof. Let $(X_n)_{n\ge 0}$ be a Markov chain with transition matrix P and $X_0 = x$, and $(Y_n)_{n \ge 0}$ be an Independent Markov chain with transition matrix P and $X_0 \sim \pi$. Define the **coupling time**

$$T := \min\{n \ge 0 : X_n = Y_n\},$$

and let

$$Z_n := \begin{cases} X_n & \text{if } n < T, \\ Y_n & \text{if } n \ge T. \end{cases}$$

That is, Z_n is the same as X_n until the coupling time T, after which it will synchronize its transitions with Y_n. Note that $Z_n \overset{d}{=} X_n$ for all n (same transitions), and that $Z_n = Y_n$ for all $n \ge T$. Now we claim that the following hold:

- **Claim 1:** $|\mathbb{P}_x(X_n = y) - \pi(y)| \le \mathbb{P}(T > n)$.
- **Claim 2:** there exist constants $\delta \in (0,1)$, $C' < \infty$, and $M \ge 1$ such that

$$\mathbb{P}(T > n) \le C'(1-\delta)^{n/M}.$$

Assuming that both these claims are true for now, this implies that

$$\sup_{x \in S} \sum_{y \in S} |\mathbb{P}(X_n = y \mid X_0 = x) - \pi(y)| \le \sup_{x \in S} \sum_{y \in S} \mathbb{P}_x(T > n) \le |S| \cdot C(1-\delta)^{n/M} \le C(1-\varepsilon)^n,$$

which proves the desired result.

Proof of Claim 1. Note that

$$\mathbb{P}_x(X_n = y) = \mathbb{P}_x(Z_n = y) = \mathbb{P}_x(Z_n = y, T \le n) + \mathbb{P}_x(Z_n = y, T > n).$$

Now

$$\mathbb{P}_x(Z_n = y, T \le n) = \mathbb{P}_x(Y_n = y, T \le n) = \underbrace{\mathbb{P}_x(Y_n = y)}_{=\pi(y)} - \mathbb{P}_x(Y_n = y, T > n).$$

The claim $\mathbb{P}_x(Y_n = y) = \pi(y)$ follows from the independence between X_n and Y_n (which implies that the conditioning on X_0 can be dropped) and since π is the stationary distribution. So,

$$\begin{aligned} \mathbb{P}_x(X_n = y) - \pi(y) &= \mathbb{P}_x(Z_n = y, T > n) - \mathbb{P}_x(Y_n = y, T > n) \\ &= \mathbb{P}_x(T > n) \cdot [\mathbb{P}_x(Z_n = y \mid T > n) - \mathbb{P}_x(Y_n = y \mid T > n)] \qquad (1) \\ &\le \mathbb{P}_x(T > n) \end{aligned}$$

since the probability of any event is between $(0, 1)$ and so the second multiple in (1) is at most one. This completes the proof of Claim 1.

Proof of Claim 2. It can be shown that there exists an integer $M < \infty$ such that $(P^M)_{xy} > 0$ for all $x, y \in S$ (Exercise! This uses irreducibility and aperiodicity.)

Let $\delta := \min_{x,y\in S}(P^M)_{xy} > 0$. Then $\mathbb{P}(T > M) \le 1 - \delta$. Indeed, by independence,

$$\begin{aligned} \mathbb{P}_x(T > M) \le \mathbb{P}(X_M \ne Y_M \mid X_0 = x) &= 1 - \sum_{y\in S} \mathbb{P}(X_M = y, Y_M = y \mid X_0 = x) \\ &= 1 - \sum_{y\in S} \underbrace{\mathbb{P}_x(X_M = y)}_{\ge\delta} \mathbb{P}(Y_M = y) \\ &\le 1 - \delta \sum_{y\in S} \mathbb{P}(Y_M = y) = 1 - \delta. \end{aligned}$$

Furthermore, by the strong Markov property, we have

$$\mathbb{P}_x(T > kM \mid T > (k-1)M) \le 1 - \delta.$$

This can also be checked by direct computation by conditioning on all the possible values that $X_{(k-1)M}$ can take and arguing similarly as above. Thus,

$$\mathbb{P}_x(T > kM) = \mathbb{P}(T > kM \mid T > (k-1)M) \cdot \mathbb{P}_x(T > (k-1)M) \le (1-\delta)^k.$$

If $n = kM$, then we have $\mathbb{P}_x(T > n) \le (1-\delta)^{n/M}$. In general, for any n, $\mathbb{P}(T > n) \le (1-\delta)^{\lfloor n/M \rfloor} \le (1-\delta)^{n/M-1}$, so the claim holds with $C' = (1-\delta)^{-1}$. $\square$

Remark 13.15 (On couplings)**.** We can define couplings of random variables on general probability spaces, which is an important concept that finds many other applications. More precisely, a **coupling** of two random variables X, Y is a pair (X', Y') defined on the same probability space such that $X' \overset{d}{=} X$ and $Y' \overset{d}{=} Y$ (i.e., with the correct marginals).

- A coupling always exists: e.g., we can choose X' and Y' to be independent copies of X and Y respectively.
- "Good couplings" are more interesting: for example, we may want to construct a coupling (X', Y') with good properties such as $X' \le Y'$ a.s.; minimizing $\mathbb{P}(X' \ne Y')$; or minimizing $\mathbb{E}[\mathrm{dist}(X', Y')]$.

An application of couplings is to (easily!) prove the following result (which is not obvious):

Lemma 13.16. *If $X \sim \mathrm{Binomial}(n, p)$ and $Y \sim \mathrm{Binomial}(m, p)$ with $m > n$, then for any z, $\mathbb{P}(X \ge z) \le \mathbb{P}(Y \ge z)$.*

Proof. Let $Z_1, \ldots, Z_n, Z_{n+1}, \ldots, Z_m$ be i.i.d. Bernoulli(p) random variables. Define $X' = Z_1 + \ldots + Z_n$ and $Y' = Z_1 + \ldots + Z_m$. By construction, $X' \stackrel{d}{=} X$ and $Y' \stackrel{d}{=} Y$. Moreover,

$$Y' = X' + \underbrace{(Z_{n+1} + \ldots + Z_m)}_{\geq 0} \geq X' \quad \text{a.s.}$$

So,

$$\mathbb{P}(Y \geq z) = \mathbb{P}(Y' \geq z) \geq \mathbb{P}(X' \geq z) = \mathbb{P}(X \geq z). \qquad \square$$

Remark 13.17 (On periodicity)**.** If $(X_n)_n$ is an irreducible Markov chain with period ℓ, then averaging over ℓ consecutive steps gives a distribution that converges to the (unique) stationary distribution π: i.e., the distribution of $(X_{\ell n} + X_{\ell n+1} + \ldots + X_{\ell n+\ell-1})/\ell$ tends to π as $n \to \infty$.

Examples of stationary distributions

(a) Two component graph.

$$\begin{tikzcd} A \arrow[d, leftrightarrow] & C \arrow[d, leftrightarrow] \\ B & D \end{tikzcd} \qquad \text{Transition matrix: } P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

There are (at least) two stationary distributions: $\pi^{(1)} = (1/2, 1/2, 0, 0)$ and $\pi^{(2)} = (0, 0, 1/2, 1/2)$. (Any convex combination $\pi = \alpha\pi^{(1)} + (1-\alpha)\pi^{(2)}$ with $\alpha \in [0,1]$ is also stationary.)

(b) Random walk on undirected graph Let $G = (V, E)$ be a undirected graph on N vertices and $|E|$ edges. Then

$$\pi(v) = \frac{\deg(v)}{2|E|}$$

is a stationary distribution. Why?

$$(\pi P)_y = \sum_x \pi_x P_{xy} = \sum_{y \sim x} \frac{\deg(x)}{2|E|} \frac{1}{\deg(x)} = \frac{\deg(y)}{2|E|} = \pi_y.$$

If G is connected (i.e., irreducible), then π is unique. Note that P does not have to be aperiodic (e.g., imagine G is an odd cycle). An easy fix: consider a **lazy random walk** – with probability $p \in (0,1)$, do nothing and stay in the current state. Exercise: check that the same stationary distribution works.

(c) Reversible Markov chains This is another important class of Markov chains:

Definition 13.18. We say that a Markov chain $(X_n)_n$ with transition matrix P is **reversible** with respect to a distribution π if it satisfies the *detailed balance equations*:

$$\pi_x P_{xy} = \pi_y P_{yx} \quad \forall x, y \in S.$$

Proposition 13.19. *Let $(X_n)_n$ be a Markov chain that is reversible with respect to a distribution π. Then π is a stationary distribution.*

Proof. Sum both sides of the detailed balance equations over x:

$$(\pi P)_y = \sum_x \pi_x P_{xy} = \sum_x \pi_y P_{yx} = \pi_y \sum_x P_{yx} = \pi_y,$$

since P is a stochastic matrix (i.e, rows sum to one). $\square$

Example: the random walk on an undirected graph $G = (V, E)$ considered above is reversible with respect to the distribution $\pi(v) = \deg(v)/2|E|$ (exercise!), which implies that π is a stationary distribution.

Interpretation: if $(X_n)_n$ is reversible with respect to π and $X_0 \sim \pi$, then the distribution of $(X_0, X_1, \ldots, X_n)$ is the same as its time-reversal $(X_n, X_{n-1}, \ldots, X_0)$. To see this, note that for any $x_0, x_1, \ldots, x_n$, reversibility implies that

$$\pi(x_0)P(x_0, x_1) \ldots P(x_{n-1}, x_n) = \pi(x_n)P(x_n, x_{n_1}) \ldots P(x_1, x_0).$$

Equivalently, we can rewrite this as

$$\mathbb{P}_\pi(X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n) = \mathbb{P}_\pi(X_0 = x_n, X_1 = x_{n-1}, \ldots, X_n = x_0).$$

(d) Simple symmetric random walk on $\mathbb{Z}$. This does not have a stationary distribution! If it exists, then from the structure of the transition matrix P, we must have $\pi_x P_{xy} = \pi_y P_{yx}$ for all $x, y \in \mathbb{Z}$. But then this implies that $\pi_x = \pi_y$ for all $x, y \in \mathbb{Z}$, which is impossible to normalize to obtain a distribution.

(e) Top to random shuffle. Consider shuffling a pack of n cards, so the state space is the set of all $n!$ permutations of the deck S_n. In particular, consider the following shuffle: at each time, take the top card and put it in a uniformly random location in the deck (it could stay at the top). Note that this defines an irreducible and aperiodic Markov chain (think this through!). Moreover, its stationary distribution is the uniform distribution π with $\pi_x = 1/n!$ for all $x \in S_n$.

Why? Let's directly check: note that $P_{xy} = 1/n$ if and only x and y have the same order, ignoring the top card of x (for each y, there are n such x), and is zero otherwise. Hence,

$$\sum_x \pi_x P_{xy} = \frac{1}{n!} \cdot n \cdot \frac{1}{n} = \frac{1}{n!} = \pi_y.$$

How long does it take for this shuffle to mix the pack of cards – that is, become "close" to the stationary distribution? More precisely, the **mixing time** is defined as the first time that the total variation distance (defined above) to stationarity, starting from the worst initial state, falls below a given tolerance, say, 1/4: $t_{\text{mix}} := \min\{t \geq 0 : \max_x \|\mu_x^{(t)} - \pi\|_{\text{TV}} < 1/4\}$. Recall from the proof of Theorem 13.13 that we considered the coupling time T of two Markov chains, X_n started from any fixed state $X_0 = x$, and Y_n with $Y_0 \sim \pi$. It can be shown that the coupling time estimates (upper bounds) the mixing time: $\|\mu_x^{(t)} - \pi\|_{\text{TV}} \leq \mathbb{P}(T > t) < 1/4$.

For simplicity, we will consider the reversed Markov chain: i.e., the random-to-top shuffle where a card is selected uniformly at random and moved to the top. It can be shown that this has the same mixing time as the top-to-random shuffle. We will consider the following coupling of X_n and Y_n: at time n, a random card is selected uniformly at random and moved to the top for both chains (possibly from different locations). Observation: once a card is selected, it remains at the same position for both x and y. Therefore, the coupling time is bounded by the coupon collector time T' (once all n different cards have been selected at least once, X_n and Y_n are identical, so the two chains must have coalesced beforehand). Hence, this implies that $t_{\mathrm{mix}} \sim n \log n$.